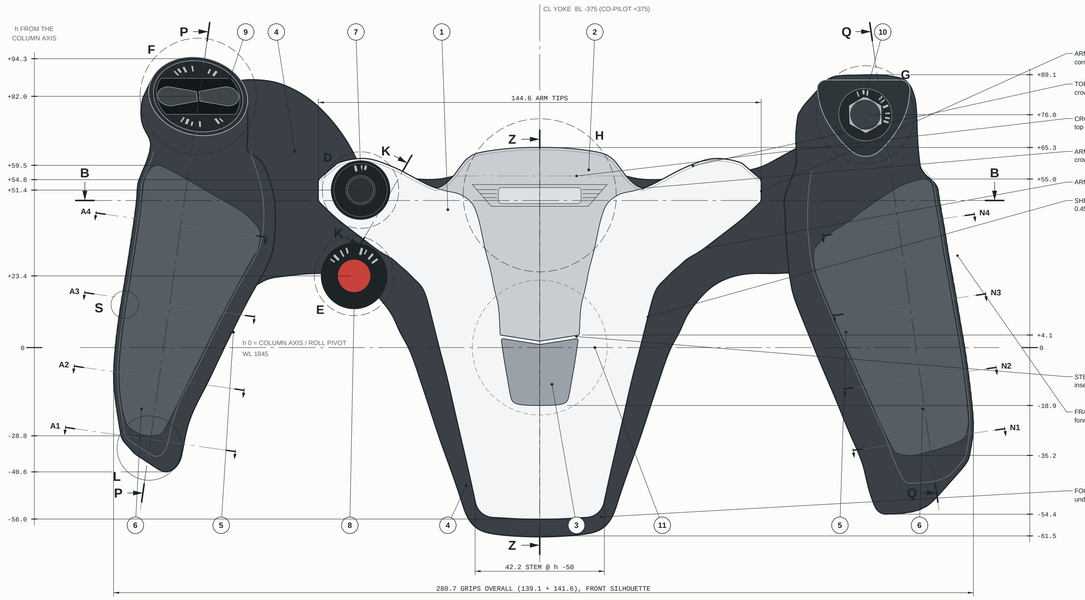
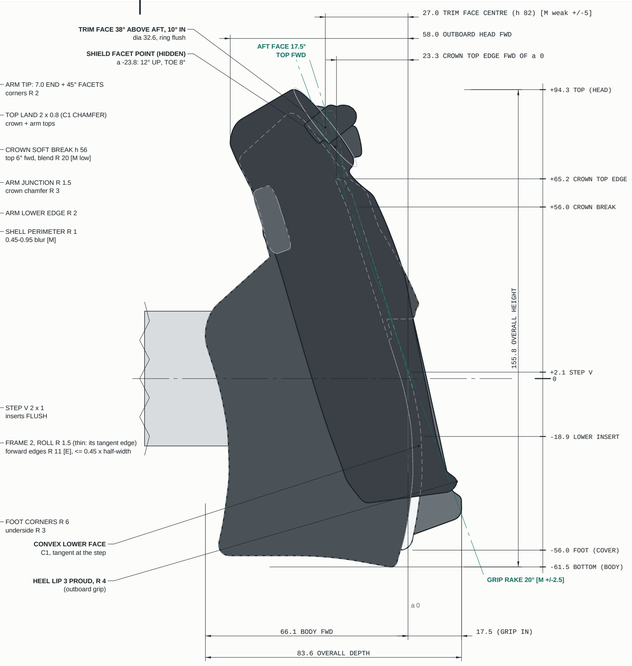


SECTIONS Z-Z, K-K, A1-A4 / N1-N4, PROFILES P-P / Q-Q, DETAILS D-H, J, L, S: SHEET L6Y2

FRONT ELEVATION - PILOT YOKE

LOOKING FORWARD, AS THE PILOT SEES IT (CO-PILOT YOKE = MIRROR IMAGE) - SCALE 1:1

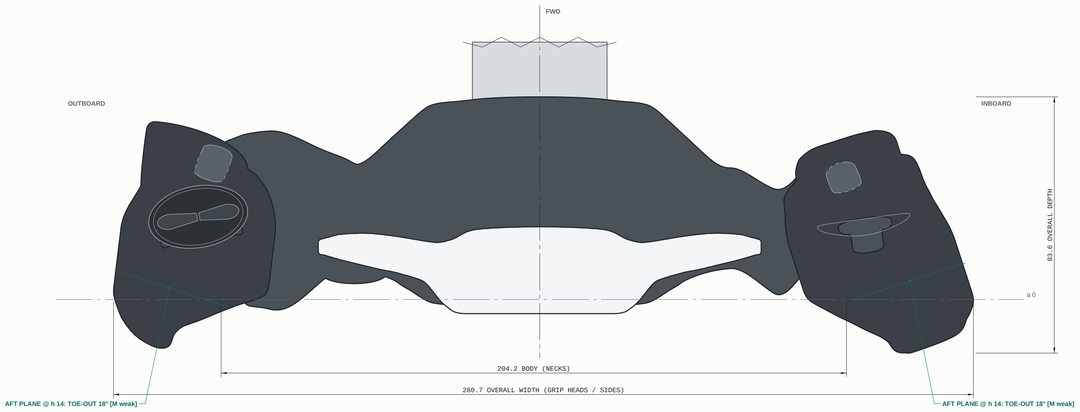


SIDE ELEVATION

FROM OUTBOARD, NOSE LEFT, NEUTRAL PITCH; a 0 = X 3700; COLUMN BROKEN (L6Y2 DETAIL J) - 1:1

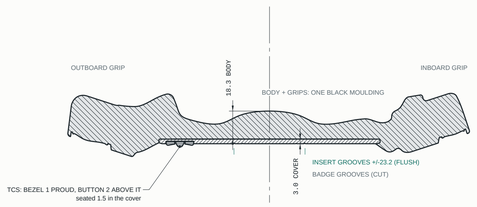
GRIP PLANES [M], 6 VIEWS INCL. P1046409: RAKE 20 +/-2.5, a0 -3.75 +/-3, DEPTH 27 +/-4, WALLS 0.6 / 0; TOE 18 WEAK +/-8 (OPEN POINT 2)

BODY (HIDDEN): TOP 42° FROM THE HORIZONTAL, TOP-TO-FRONT R 6, DEPTH 60 [E], FOOT R 5 [E] (L6Y2 Z-Z)



VIEW FROM ABOVE

FORWARD UP, PILOT YOKE - SCALE 1:1; GRIP PLANES ON 6 VIEWS (L6Y2 PHOTO CHECK); TOE WEAK (OPEN POINT 2)



SECTION B-B

h 48 (BADGE / TCS), EXACT CUT OF THE SOLIDS, FROM ABOVE - SCALE 1:2

ITEMS & FINISHES (render/lookdev names; 'new' = to add)

#	PART	MATERIAL	FINISH / COLOUR [SOURCE]
1	WHITE COVER (Y SHELL)	yoke_white	gloss clear-coat white, neutral-cool (0.77 / 0.78 / 0.80), fine pearl; R 1 edges, top land [M]
2	UPPER INSERT = CROWN, BADGE	yoke_silver	COOL satin-aluminium 0.58 / 0.61 / 0.67, metallic 0.85, rough 0.34 (rev Q: cooler and darker than the cover); flush, 0.6 x 0.5 grooves; the badge's blank plate 0.2 proud [M / E]
3	LOWER INSERT	yoke_silver_dark	cool smoke silver 0.29 / 0.30 / 0.33, metallic 0.85, rough 0.36 (rev Q); flush [M]
4	BLACK BODY (RIM, WEB, NECK)	grip_black	satin black, rough 0.42: crisp edge highlights; R 2 aft edge [M]
5	GRIP (2) WITH FRAME, HEEL LIP	grip_black	satin black; 2.0 frame R 1.5, heel lip 3 proud R 4 [M]
6	LEATHER PAD (2)	yoke_leather	charcoal 0.05, rough 0.58, 1.1 pebble grain, ~1 creases, 2.5 below the frame, groove 1.0 x 0.5 + bead 0.4 at its edge [M / E]
7	TCS BUTTON	gloss_black / black	gloss black bezel 1 proud [E], satin domed button 2 proud of it [M low]; legend blocks (legend_grey)
8	AFCS DISCONNECT	button_red / grip_black	deep red-cap 0.50 / 0.004 / 0.012, clear coat 0.15 (rev Q), NOT emissive, on a boss in the body's satin black (rev Q) [M]; legend blocks
9	PITCH TRIM ON ROUND FACE	yoke_trim_face / yoke_rocker / yoke_slot	rev P: SATIN CHARCOAL ring + lunes 0.04, rough 0.45 (as light as the body or lighter); rocker near-black satin 0.02, rough 0.30, ends 9.5 proud, crowned; 2.5 deep near-black MATTE slot 0.01, rough 0.8 (the darkest part) [M / E]; legends
10	PUSH BUTTON ON SHIELD	gloss_black / yoke_nickel / black	rev O: GLOSS-BLACK legend disc 17.5, 0.75 proud, legend blocks at its rim; small SATIN NICKEL hex 12 A/C (warm); smooth black cap, centre dimple [M]; function: MAP LIGHTS (no lettering)
11	COLUMN	yoke_column (new)	satin dark grey tube 0.04, metallic 0.5, rough 0.4 (rev Q), dia 44 [E] (never seen in a PRO photo); the viewer parks the free yokes 35 forward (YOKE rest, rev P), the elevators trailing-edge down with them (rev Q)
12	COLUMN SURROUND (PANEL)	pcl_pewter / panel_dark	satin pewter J band 22 wide, both ends round (rev P), charcoal dished recess [M / E]
13	FORWARD-FACE SWITCHES (2) [E]	grip_black	PTT trigger (outboard grip), checklist rocker (inboard): placeholders, never seen (L6Y2 detail W) [E]
14	LEGENDS	legend_grey (new)	words of plain letter blocks 0.22 of the pitch, uneven widths (rev P), set with uneven gaps and heights (rev Q), tone ~0.40, 0.12 proud – NO lettering (rev N); the pilot's in reading order, not mirrored
15	COVER GROOVES / STEP	yoke_groove (new)	rev Q: the V flanks of the parting grooves, the step's down-facing flank and end cones, the badge outline / speed lines dark satin 0.03, rough 0.6 (their cavity shadow: dark lines in every photo); the step's lower flank keeps the cover white (the lit line under it) [M]

MAIN DIMENSIONS (read from the table / the solids)

DIMENSION (mm, YOKE-LOCAL)	VALUE	SOURCE
shell height / about h 0	121.3 / -56.0, +65.3	[M] rev J landmark-pose frame
arm tips / stem at h 0 / at h -50	144.6 / 65.6 / 42.2	[M] sigma 1.5
crown above the arm junctions	13.3	[M] parting grooves
step / lower insert bottom	h 2.1 (V) / -18.9	[M]
black rim outboard of the stem at h 10 / 0 / -50 / under the foot	12.6 / 10.1 / 4.5 / 5.8	[M] body - shell outlines
aft face above the step	17.5° top fwd	[M] bundle adj., +/-4
body depth / top / cover	60 / 42° from horiz. / 3	[E] / [M lower bound] / [E]
heads top: FRONT SILHOUETTE out / in	94.3 / 89.1	[derived: solids from M / E]
grip bottoms (front silhouette) out / in	-40.6 / -54.4	[derived: solids, rake / toe M, depth M / E]
grips overall: front silhouette (out + in)	280.7 (139.1 + 141.6)	[derived: solids]
CONSTRUCTION G (prism directrix): head tops / span	97.4 / 90.1; 281.3	[M] table; NOT the part (head solids trim it)
rake / toe-out / a0 @ h 14: out; in	20° / 18° / -3.75; 20° / 18° / -6	[M] 6 views incl. P1046409: +/-2.5° / toe weak / +/-3
walls (0 = aft normal, 1 = fore-aft) out / in	0.6 / 0	[M] +/-0.2
grip depth (normal) bottom / mid / head, out; in	27.5 / 27 / 27; 27.5 / 27 / 27	[M] P1046409 slot wall +/-4; in = out [E]
pad out: top / slot corner / bottom	59.5 / (91, 38) / -28.8	[M] diagonal top 31°
pad in: top / slot corner / bottom	55.0 / (89.4, 32) / -35.2	[M] diagonal top 33°
frame / leather below it / heel lip	2 / 2.5 / 3 proud R 4	[M]
trim face centre (u, h, a) / dia / normal	(111.5, 82, -27) / 32.6 / 38° up, 10° in	[M] (u, h) 7 views; a [M weak +/-5]
shield facet point (u, h, a) / normal / size	(106, 75.6, -23.8) / 12° up, toe 8° / 31 x 25.5	[M] (inboard: -u)
switch axes (u, h, a): button / TCS / AFCS	(106.3, 76.0, -23.8) along the facet normal / (58.5, 51.4, -15.5) / (60.6, 23.4, -8.2) along the 17.5° face's normal	[M]; TCS on the face, AFCS boss face = face - 3 + 1.5
TCS / AFCS cap / boss	(58.5, 51.4) 19.2 / 10.8 x 5.5 / 21.6	[M]
forward-face switches out / in (h, L x W, proud)	h 56, 22 x 10, 4 / h 50, 18 x 10, 3.5	[E] placeholders
column dia / axis	44 [E] / h 0 = WL 1845	kept; surround re-centred on it (owner 2026-10-11)
pitch travel push / pull / roll	65 / 90 [E] / +/-70 [E]	push: owner 2026-10-11 (was 90 [E])

KNEES vs YOKE (L6B), REV H > I > J > K (RE-SWEPT) > M - CRITERIA UNCHANGED, ACCEPTED AS MEASURED

OCCUPANT	LEVEL mm	ROLL+/-70 mm	CONTACT deg	KEEP 25 deg	RUDDER mm
95th male	38 > 35 > 40 > 25 > 24	-19 > -18 > -19 > -19 > -19	17.5 > 17.5 > 15.0 > 12.5 > 12.5	5.0 > 2.5 > 5.0 > 0.0 > -1.0	-11 > -13 > -9 > -22 > -24
50th male	50 > 48 > 52 > 38 > 36	-13 > -14 > -14 > -14 > -14	25.0 > 25.0 > 22.5 > 22.5 > 22.5	10.0 > 10.0 > 10.0 > 5.0 > 5.0	5 > 3 > 7 > -7 > -8
5th male	51 > 48 > 51 > 37 > 35	-21 > -21 > -22 > -22 > -21	22.5 > 22.5 > 22.5 > 17.5 > 17.5	10.0 > 10.0 > 10.0 > 5.0 > 5.0	20 > 18 > 22 > 8 > 7
5th female	76 > 78 > 81 > 67 > 65	5 > 0 > 0 > 0 > 0	87.5 > 87.5 > 87.5 > 87.5 > 87.5	27.5 > 25.0 > 25.0 > 20.0 > 20.0	55 > 53 > 56 > 42 > 40

NOTES

1. OWNER 2026-10-05: L6Y rev D / L6Y2 rev C APPROVED as the basis of the 3-D yoke; this rev E carries the judges' round-4 drawing-stage cleanup (the measured outline unchanged but the outboard grip's heel corner and waist, FID4-01 / 02). Signed off with the approval: the rev L hidden estimates foot round R 14 -> R 5 under a 4 mm draft cap and inboard head cut R 3 -> R 1.5.
2. Yoke-local mm: u outboard (pilot- -y), h up from the hub centre = the column axis / roll pivot (WL 1845, BL -/+375), a aft of the aft face at the insert step (X 3700 at neutral pitch). Drawn from models\interior.py YOKE_DESIGN rev M through the solids of model\yoke_geom.py; L6 / L6B take the same solids (yoke_outline_yz, yoke_sweep_pieces).
3. CURVE LAW (table curve_law, rev M text): each outline split at its corner samples into runs, each run a C2 cubic spline on centripetal knots ('natural' ends at the corners, periodic for a closed outline without corners), sampled every 0.05 (arcs too); corners R 0 sharp, R > 0 a tangent fillet (cubic Bezier arc, tangent length R tan(θ/2)) capped at 0.45 of the shorter run: none capped in rev M); the lower aft face a C1 Hermite tangent to the 17.5° plane. The solids use the EXACT distance to those polylines (no raster; beyond 0.3 mm a C2 B-spline of exact node distances); views are exact silhouettes, sections exact cuts.
4. SURFACES: the cover's face (the 17.5° plane, the lower Hermite, the crown bevel) trimmed by its outline; the body: sides / bottom = its outline without the top edge, top = a plane 42° from the horizontal through the outline's top edge (body_top, a C2 cubic in (u)), forward face, rounds in order R 2 / R 2 / R 8 (R 5 at the foot, the draft's offset capped at 4 there; the crown's top edge h 65.2 = the outline's 65.3 less its R 1); the web's aft face = the trim face's plane 3.5 below it (web); each grip: its AFT SIDE = the prism's aft plane trimmed 3 inside the head's shaping surface above the plane height 67 / 61, drafted outward below it from a bend of R 4 (an exact 2-D distance), cut edge R 1.5, joined R 3 to the HEAD's aft shape (round face on its flared cylinder / shield facet + fall-off, x chamfer R 3, floor R 3); then the walls (outline G, a construction curve) and the forward face cut both, edges R 1.5 easing to R 2 / R 11 (capped) easing to R 3 over the head, the head's top onto the forward face R 3; body + grips one moulding joined by a ROLLING-BALL R 3 (a true round where they meet at an angle, none where they run flush); the pad pocket, groove and bead cut after that join. Every edge is a tabled round or a tabled crease (L6Y2 CORNERS & RADII).
5. STAGE-3 SURFACE ROUTE: each named surface a grid_surface over its parametrisation (planes, the face Hermite, the head cylinder about the face normal) trimmed by cad.mesh.trim with these exact 2-D fields; each tabled round a sweep_profile of its radius; the head-zone trimming surface is a draft from the head surface's offset curve; vertex normals from the analytic surfaces. The solids stay the inside / outside oracle. Acceptance (model\yoke_checks.py, L6Y2): vertex normals within 2° at the 95th percentile on 9 patches; NO UNTABLED CREASE in an adaptive scan of 14 views (every pixel pair turning > 4° refined 256x); the joins read smooth fields (no ridge at their rounds' own surface points); each pad's exposed surface = its drawn section; no untabled section cusps; one piece per side in every h section above 62; nothing above the trim face / shield.
6. White moulding symmetric; the black body and the grips are NOT: an outboard web running into the trim head's rim (10-11 o'clock), an inboard LOW NECK with an open notch under the inboard head; switches outboard (TCS, AFCS, trim), the MAP LIGHTS push button on the inboard head; co-pilot yoke = mirror image.
7. NO MARKINGS: the badge is an abstract winged plate with a blank raised field, every legend plain letter blocks.
8. Not modelled: the rev H column bellows (none in 8 photos), the PC-24's grip tab (not on the PRO). The grips' FORWARD (finger-side) faces are UNVERIFIED -- no photo shows them; AOPA (2025-03-17) lists push-to-talk and checklist among the yoke's one-hand functions and only trim, TCS, AFCS DISC and MAP LIGHTS are seen: drawn as [E] PLACEHOLDERS (rev M, judges FID4-08): a PTT trigger on the outboard grip, a checklist rocker on the inboard grip (item 13, L6Y2 detail W; open point 7).
9. GRIPS' 3-D LIE (L6Y2 photo check): rake / toe / a0 / walls / depth fitted per grip to the curated silhouettes of 6 views -- P1046408 L / R, P1046410, Mexico L / R and P1046409 (posed with a two-view camera, the co-pilot yoke 25° from inboard) -- with the front outline held: rake 20 +/-2.5, a0 -3.75 +/-3, walls 0.6 / 0 +/-0.2, depth 27 +/-4 (P1046409's slot wall; no bottom taper); the toe is weakly determined by views from inboard only (open point 2). The AOPA photo is not used: its co-pilot camera is EXTRAPOLATED from the pilot-yoke solution (judges FID4-05; re-calibrating it on the panel round the co-pilot yoke is open) -- until then the toe, the depth law and the head-depths stay [M weak].
10. Knee checks on the rev M solids: the 95th clears the yoke by 23.8 mm wings level (criterion 25 kept: short by 1.2), roll to contact 12.5°, keeping 25 mm to 0.0° -- ACCEPTED AS MEASURED (owner 2026-10-05; L6B open point 1 closed). K column = rev K re-swept: as issued (37 mm, 15°) the sweep had dropped a band of the inboard grip (a doubled vertex in the band clip, fixed in rev L).

OPEN POINTS FOR THE OWNER

1. Fore-aft station: EVERY posed photo puts the aft face forward of the drawn neutral -- P1046408 -60 / -55, P1046410 -65, P1046409 -37 (s/n 3001, gust lock on: 'CONTROLS LOCKED'), Mexico -38 / -34 and the AOPA photo -14 / -26 (no lock placard) [+/-20-30 each along the line of sight]. Unlocked, a parked yoke rests forward: the elevator droops trailing-edge down under its weight [E: physics, no PC-12 source]. Option: neutral 45 further forward (hub_dx 140 -> 95): the 95th's contact roll 12.5° -> 12.5°, wings-level clearance 23.8 -> 23.8 mm (L6B sweep: the knees lie inside the +/-90 pitch-swept slab either way, so the station does not move the knee rows). Kept 140 (owner); rev P: the viewer parks the free yokes at the unlocked photos' rest, 35 forward (YOKE rest; 3-D judges r3 FID3-02).
2. Grip toe: all six usable views look from INBOARD (15-27°): the silhouettes prefer 7 (outboard, +5 % at 18) and 16.7 (inboard), the inboard PAD traces alone ~26 (judges FID4-06: both optima); the one view from outboard (AOPA, its camera extrapolated from the pilot-yoke solution, FID4-05) favours 18-26, and there the outboard pad's lower outer edge reaches the drawn grip silhouette (FID4-07: leather wrapping the outer edge low on the grip, or an outer frame + roll under 2.0 < R 1.57) -- to decide after that camera's re-calibration. Kept 18 / 18. The trim head's depth is weak too: a -27 vs the judges' fit (110.7, 83.7, -23.6), P1046409 / 10 agreeing in sign (FID4-04).
3. Column: CLOSED (owner 2026-10-11) -- the J band and the U recess re-centred on the column axis / roll pivot (hub WL 1845; the photos' joint line reads 1821 +/-10): recess h -48 .. 27 about it, the band 1.0 off the column; push travel 65 (was 90 [E]), pull 90 (L6Y2 detail J). Still open: a J band or a full collar round the column.
4. Shell skew (judges FID4-03): ONE left / right pattern in all 8 views of both yokes, each in its own sense (means of 8, + = photo outside): stem -1.5 / +1.7 (~1.6 inboard), arm tips +1.7 / -1.5 (both ~1.6 outboard), inboard arm top -1.7 / lower edge +1.7 (~1.7 low); crown, chamfers, foot within 1.2. Kept SYMMETRIC: one decision for the stem, both tips and the inboard arm.
5. Knees: CLOSED (owner 2026-10-05) -- the 95th's 23.8 mm wings level (criterion 25 kept) and contact from 12.5° are accepted as measured.
6. Crown soft break (R 20 [E]) vs a crease; hidden: body depth 60, the necks' depth 30, rocker / cap heights.
7. Forward-face switches (PTT, checklist): drawn as [E] placeholders (item 13, L6Y2 detail W) -- a photo of a grip's finger side would replace them.

REVISIONS - PC12-L6Y

REV	DATE	DESCRIPTION
A	2026-10-05	first issue (owner 2026-10-05: 'the yoke isn't quite right especially if you look at the details'): yoke re-measured on 8 PRO photos; YOKE_DESIGN rev I replaces the rev H outline
B	2026-10-05	rev J (judges r1): the yoke as solids (model\yoke_geom.py) -- every view / section exact; curve law + corner tables; one body with an outboard web and an inboard LOW NECK (open notch under the head); pads with diagonal tops in a raised 2 mm frame, rolled heel lip; grips refitted per grip (toe 18 / 18) on 4 pad views; ROUND trim face and SHIELD facet heads, triangulated switch centres; crown / arm tops 1.7 lower (landmark-pose frame), top land, crown crease; body top falling forward 35 deg; inserts flush; AFCS round boss, hex button collar, mirrored rocker lobes; sheet L6Y2 added (sections, heads, switches, column, photo check)
C	2026-10-05	rev K (judges r2): exact 2-D distance fields (no raster), Stage-3 surface route + acceptance checks; grip walls fore-aft (outboard 0.6) -- slot-side and heel silhouettes on the curated photo data; forward-edge R capped, PCHIP depth, heel lip R 4, pad groove + bead built; body top 42 deg from horizontal, top-to-front R 8; the web held under the trim face; AFCS boss, TCS, rocker, hex button as solids; TCS button proud; crown soft break R 20; grip planes JOINT-FITTED to 7 views (5 confirm the table +/-3 deg / +/-3-5 mm; the AOPA photo disagrees: open); outboard heel corner pulled in; ordinate dimensions, cutting planes, detail circles, all values computed
D	2026-10-05	rev L (judges r3): P1046409 posed (two-view camera) -- grip planes re-fitted on 6 views, depth 27 from its slot wall (no bottom taper), PROVISIONAL removed, the AOPA photo dropped; the web runs into the trim head's rim (no fin) on the face's plane; head zone of named surfaces and tabled rounds; body top = a plane through the outline's top edge, rounds in order; C2 curve law; U column recess [M]; forward-face switches unverified (PTT expected); knee sweep with the inboard grip's lowest band restored (rev K re-swept); acceptance clean: the head zone's draft starts C1, inboard cut R 1.5, foot R 5 under a 4 mm draft cap, unit-gradient top / aft
E	2026-10-05	rev M (judges r4, owner: rev D approved as the 3-D basis, the 95th knee accepted as measured): pad pocket cut after the body-grip round (the outboard pad's slot-side corner no longer buried); the moulding joined by a rolling-ball R 3 (no bulge where flush); head zones: head and prism share walls and forward face, joined R 3 (was 1), trim inset 3, draft start R 4, outboard hz 67; body top line and heel lip C2; outboard heel corner raised 2.5 (R 9), waist 127.4 at h 66; [E] forward-face switches; heads' a and switch axes dimensioned; adaptive crease / field-ridge / pad-exposure acceptance, and what it found fixed: the head's top onto the forward face R 3, body sides x top a rolling-ball R 2 between pieces rounded onto the forward face (aft face cut last), forward-edge radius cap C2, the inboard fall-off's directrix one tangent curve (R 4 withdrawn), outlines sampled 0.05, C2 fields
F	2026-10-11	rev N (3-D judges r1): legends as words of plain letter blocks (the tick rows read as a dial); trim rocker in a 2.5 deep slot (7 deg draft), satin dark-grey lobes, gloss lunes; badge: a raised blank plate 0.3 proud, push button hex collar satin nickel; column satin black (non-metallic); column recess up to h 27 (the column, about the hub, 5 mm clear), J band 22 (1 mm clear of the column), the band pewter, the recess charcoal; items 9-14 re-written. OWNER DECISIONS 2026-10-11: the J band and the U recess re-centred on the column axis (hub WL 1845; the photos read 1821), push travel 65 (was 90 [E]), pull 90; the pilot yoke's legends keep their reading order (item 14)
G	2026-10-11	rev O (3-D judges r2): item 10 -- the push button a GLOSS-BLACK LEGEND DISC 17.5 (0.75 proud) with MAP LIGHTS at its rim, a small satin-nickel hex 12 A/C under the 10.2 cap (rev N's 17.3 collar read as a big bright nut where every photo shows a dark disc); item 9 -- rocker ends 9.5 proud (6.5), lobes crowned 0.6, satin black; legend blocks 0.35 of the pitch, tone 0.45; badge: ONE outline groove (the inner border dropped), speed lines into it, plate 0.2 proud; open point 3 closed (owner 2026-10-11)
H	2026-10-11	rev P (3-D judges r3): item 9 -- the legend ring / lunes a SATIN CHARCOAL face and the rocker slot a near-black MATTE pocket (rev N / O's gloss lunes over a satin slot read inverted against every photo); legend letters 0.22 of the pitch, uneven (+/-35 %), tone 0.40; item 10 hex 12 A/C (text); item 11 column matte near-black; item 12 the J band's hook ends round; the viewer's parked yokes at the photos' free rest, 35 forward (YOKE rest); item 8 text: the rev O red
I	2026-10-12	rev Q (3-D judges r4): item 15 (new) -- the cover's groove flanks DARK (yoke_groove: the step's down-facing flank and end cones, the parting grooves, the badge outline / speed lines; the step's lit lower flank keeps the white), FID4-01; items 2 / 3 the inserts a COOL satin metal (FID4-02); item 8 the AFCS boss in the body's satin black (FID4-06); item 11 the column a satin dark-grey tube, the parked elevators coupled to the parked yokes (FID4-05, CR4-05); item 14 the legend letters set with uneven gaps and heights (CR4-06)

PILATUS PC-12 PRO

PC-24-STYLE CONTROL YOKE - VIEWS

FIRST ANGLE PROJECTION

DWG NO	REV	SCALE	SHEET	UNITS
PC12-L6Y	I	AS SHOWN	A1 · 1 OF 2	mm

DRAWN	DATE
Claude - loftkit parametric model	2026-10-12

RECONSTRUCTION FROM PUBLISHED DATA - NOT FOR CONSTRUCTION